

Pellic Vertisols:

Pellic Vertisols are heavy, clay-rich soils that develop strong shrink-swell behaviour (vertic properties) and commonly have a **dark, nearly black surface horizon**.

They are called “**Pellic**” because of that dark-coloured topsoil (often low moist chroma and sometimes higher organic matter in the surface), which distinguishes them from the redder/chromatic Vertisols.

Pellic Vertisols commonly form in low-lying or poorly drained flat plains and depressions where organic matter accumulates in the surface and periodic waterlogging occurs.

Drainage:

- **Wet season:** Poor to imperfect drainage in lowland positions; surface water may stagnate because the dense, high-clay subsoil slows vertical percolation.
- **Dry season:** Soils develop wide, deep cracks that increase infiltration locally but cause highly uneven wetting when rains return. These cracks also periodically shear roots and disrupt continuity of the rooting zone.
- **Landscape:** Typically in level to gently undulating landscapes — valley bottoms, depressions and plain positions where seasonal waterlogging and organic matter accumulation favour a dark topsoil.

Depth:

- Usually **very deep** profiles (>100–150 cm) with a thick, active vertic horizon; the effective rooting depth may be periodically interrupted by the cracking/shearing processes but physical depth is large.

Workability:

- **When wet:** Very sticky, plastic and smears easily — tillage and machinery operations in the wet season often cause structural damage and compaction of surface layers.
- **When dry:** The soil dries to extremely hard clods and crusts (rock-like), making ploughing and seedbed preparation difficult without appropriate equipment.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** High, typically large values (often 30–80 cmol(+)/kg) because the fine clay fraction (smectitic minerals) provides a large exchange capacity.
- **Base saturation:** Frequently high where parent material supplies bases; many Vertisols have good base status compared with highly leached tropical soils.
- **Nutrients:** Despite high CEC, plant-available N and P are commonly limiting — N is low because organic matter turnover may be limited and P is often fixed by clay/Fe/Al compounds. Micronutrients (Zn) can be limited under intensive cropping.

pH: Generally **neutral to alkaline** (commonly ~6.5–8.5), depending on parent material and local hydrology; acidity is uncommon in typical Pellic Vertisols.

Management:

- **Drainage & field layout:** Use surface drainage, broad beds/furrows, raised beds or ridges to move excess water away from crop root zones in the wet season and to avoid prolonged saturation in depressions.
- **Tillage timing:** Restrict tillage to the narrow moisture window (slightly moist — not sticky, not rock-hard). Controlled traffic and light, shallow cultivations reduce structural damage.
- **Soil organic matter:** Incorporate organic amendments (manure, compost, residues) to improve aggregation, reduce surface crusting and increase resilience to cracking. Pellic surfaces often have higher surface OM than other Vertisols but still benefit from additions.
- **Fertilisation:** Apply **N and P** based on soil tests. Because P fixation can be high, combine **slow-release rock phosphate** (for long-term P building) with smaller, soluble P applications (e.g., single superphosphate) for immediate crop response; split N applications reduce leaching and improve uptake.
- **Erosion & microtopography:** On level Vertisols, gilgai microrelief (micro-basins and mounds) is common — maintain vegetation strips and avoid removing natural micro-relief that helps redistribution of water.
- **Machinery & traffic:** Avoid heavy machinery when soils are wet to prevent smearing and long-term structural damage. Use appropriate implements only in the correct moisture window.

Suitable crops:

Pellic Vertisols have **high yield potential** when managed well. Common and successful crops include **cotton, sorghum, wheat, barley, teff (in Ethiopia), chickpea, field peas, and sugarcane**; with irrigation and careful management they also support rice and other cereals/forage crops. Choice depends on local hydrology and whether waterlogging can be controlled.